Math 445/545, Midterm I.

REMINDER: For a correct answer with no explanation you can get 0 points.

0. Write your name here:

1. Let R be a ring and let $a \in R$. Prove that $S = \{x \in R \mid ax = xa\}$ is a subring of R.

Solution: We need to check that S with operations inherited from R is a ring. Most importantly we need to check that S is closed under the operations. Let $x, y \in S$. By definition this means $ax = xa$ and $ay = ya$. Hence $a(x+y) = ax + ay = xa + ya = (x+y)a$, so $x + y \in S$. Also $a(xy) = (ax)y = (ax)y = (xa)y = x(ay) = x(ya) = (xy)a$, so $xy \in S$. Thus S is closed under addition and multiplication.

We need to check that S is abelian group. Clearly, addition in S is associative and commutative (since addition is associative and commutative in a bigger set R). Also S contains zero element since $a \cdot 0 = 0 = 0 \cdot a$. Finally if $x \in S$, then $(-x) \in S$ since $a(-x) = -ax = -xa = (-x)a$. Thus S is abelian subgroup of R (with respect to operation of addition).

All the remaining axioms (namely, associativity of multiplication and distributivity) of the ring hold in S since they hold in a bigger set R. Thus S is a subring of R.

2. Find a polynomial of positive degree in $\mathbb{Z}_4[x]$ which is a unit of this ring. (Hint: thinking about nilpotent elements might help)

Solution: We have $2 \in \mathbb{Z}_4$ is a nilpotent since $2 \cdot 2 = 0 \in \mathbb{Z}_4$. Hence polynomial $2x \in \mathbb{Z}_4[x]$ is also a nilpotent. Hence $1+2x \in \mathbb{Z}_4[x]$ is invertible: $(1+2x)(1+2x) = 1+4x+4x^2 = 1$ (one can also use one of the practice problems: unit plus a nilpotent is a unit).

Answer: one such polynomial is $1 + 2x$.

Comment: there are many other polynomials like this, e.g. $3 + 2x^2 + 2x^5$. Can you describe all of them?

3. Find a nontrivial idempotent (so this idempotent is not 0 or 1) in the ring $\mathbb{Z}_{143}$.

Solution: We need to find $x \in \mathbb{Z}_{143}$ such that $x^2 = x$ and $x \neq 0, 1$. Equivalently, we need a number $x \in \mathbb{Z}$ such that $x^2 - x = x(x-1)$ is divisible by $143 = 11 \cdot 13$. This will be the case if x is divisible by 13 and $x - 1$ is divisible by 11. Now just check numbers divisible by 13 as candidates for x: $x = 13, 26, 39, 52, 65$ are not working but $x = 78$ is Ok since $x - 1 = 77 = 7 \cdot 11$.

Answer: $x = 78 \in \mathbb{Z}_{143}$ is an idempotent.

Comment: there is one more solution $x = 66$. Prove that there are np other ones!

4. Let $f(x) \in \mathbb{Q}[x]$. Prove that the polynomial $f(x)^2 - 2$ has no factors (with rational coefficients) of degree 1.

Solution: A polynomial has a linear factor if and only if it has a zero. Assume that $f(x)^2 - 2$ has a zero $\alpha \in \mathbb{Q}$. Then $f(\alpha)^2 = 2$ and $f(\alpha) \in \mathbb{Q}$ (since $f(x) \in \mathbb{Q}[x]$). This is a contradiction with irrationality of $\sqrt{2}$.

5. (a) Show that matrices of the form $\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}$ where $a, b \in \mathbb{R}$ with usual operations of matrix addition and multiplication form a ring.

Solution: Let R be the set of matrix above. We have

$$\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \pm \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a \pm c & b \pm d \\ 0 & 0 \end{pmatrix},$$

so R is closed under addition and subtraction; thus R is an abelian group.

Similarly,

$$\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} ac & ad \\ 0 & 0 \end{pmatrix},$$

so R is closed under multiplication. The associativity and distributivity are obvious since they hold for the matrix addition and multiplication. Thus R is a ring.

(b) What is the identity element in the ring from part (a)?

Solution: We have

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Thus the ring R has no identity.

Answer: this ring has no identity element.